

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	BALA S
Register Number	192524427

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I BALA S (192524427) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: bala

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1:

AIM:

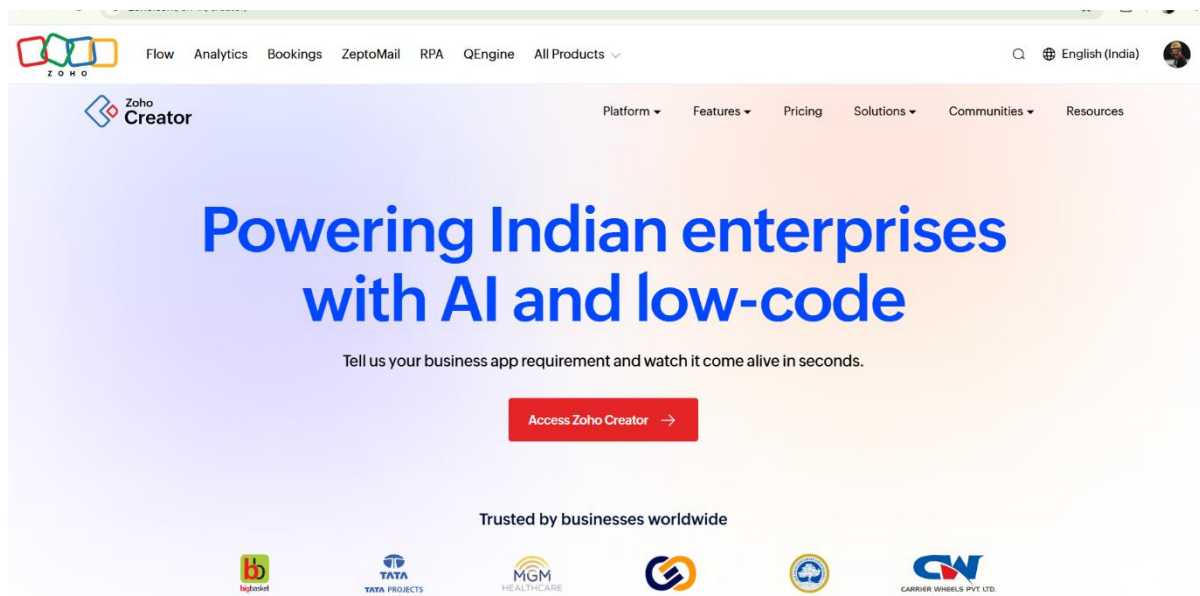
Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

1. Go to zoho.com.
2. Select one application step.
3. Enter application name as library book reservation system.
4. Created new application as library book reservation system.
5. Select one form
6. The software has been created.

IMPLEMENTATION:

1. GO TO ZOHO.COM



2. SELECT ONE APPLICATION



3. ENTER APPLICATION NAME



Create from scratch

Application Name *

Enter your application name

☐ Enable Environment [Learn More](#)

Create Application

4.CREATE NEW APPLICATION

LIBRARY MANAGEMENT
STUDENT DETAILS

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123
Number

15
Date

Time

Drop Down

123 STUDENT ID

STUDENT NAME

DEPARTMENT

YEAR

Email

Phone

ADDRESS

Field Properties

Field name

ADDRESS

Field link name

ADDRESS

[View Field References](#)

Choices [Advanced](#)

☒ Renaming the choices will update the values in the existing records, if any.

☐ Choice 1

☐ Choice 2

☐ Choice 3

☐ Alphabetical Order

Validation

5. SELECT ONE FORM

How would you like to create your form?

On my own

+

From scratch

Import Schema

Using Zia

From a template

Cab Booking

Conf Room Booking

Appointment Booking

Hotel Room Booking

Complaints

50 +
View More

6.THE SOFTWARE HASE BEEN CREATED.

SIMATS Library Book Re...
BOOK RESERVATION

BOOK RESERVATION

BOOK RESERVATION Report

BOOK RESERVATION

Student Name

Registration Number

Department

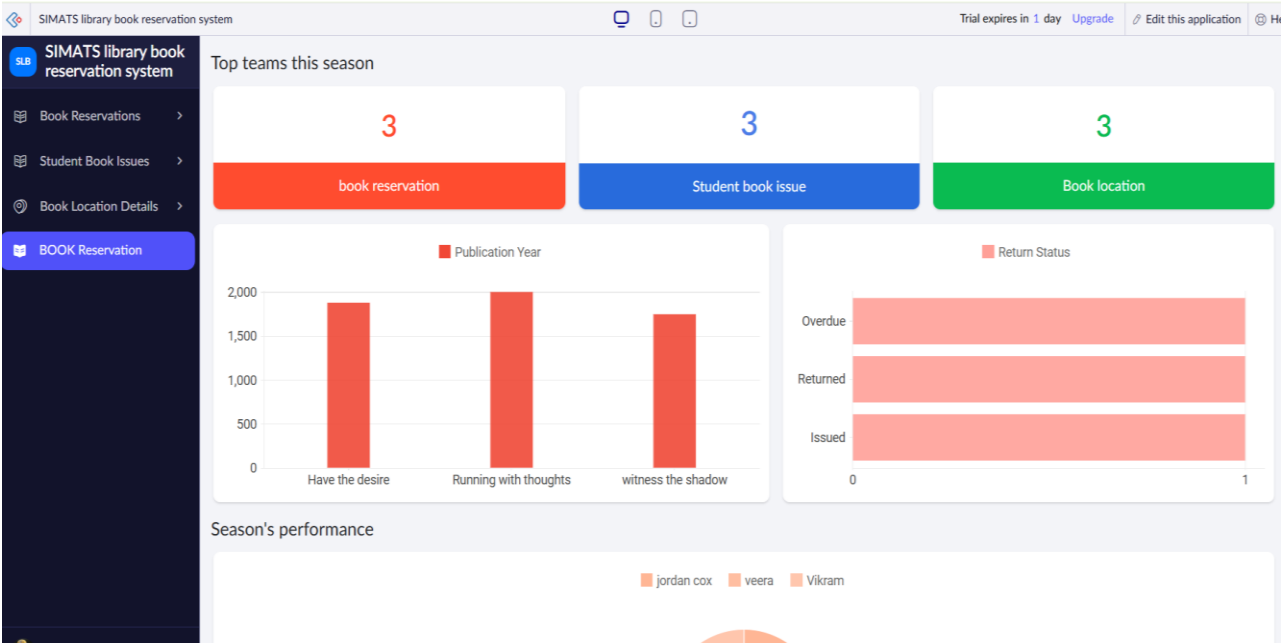
Email

Phone

Book Title

Author Name

ISBN Number



Experiment 2:

AIM:

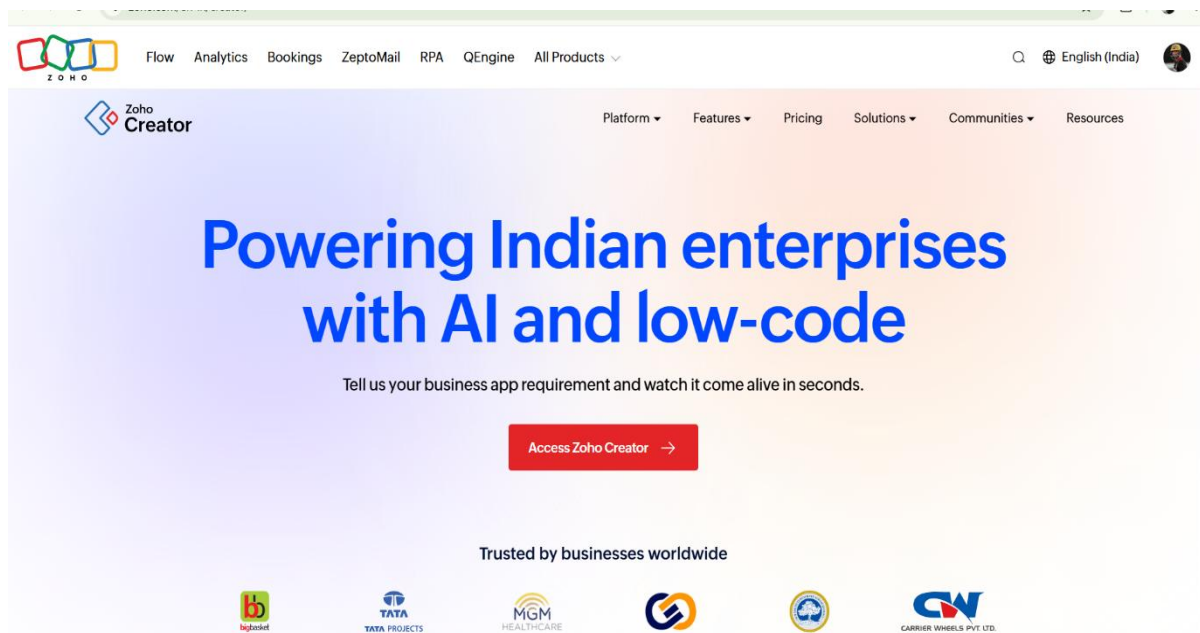
Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

1. Go to zoho.com.
2. Log in to your Zoho account.
3. Open Zoho Creator from the Zoho applications.
4. Click Create New Application.
5. Enter the application name as Payroll Processing System.
6. Create a new application.
7. Create a new form named Employee Payroll Details.
8. The Payroll Processing System software has been successfully created and deployed on the Zoho cloud platform.

IMPLEMENTATION:

1.GO TO ZOHO.COM



2.LOGINTOTHE ZOHOCOM.

Create from scratch

Application Name *

Payroll Processing System

☐ Enable Environment [Learn More](#)

Create Application

3.SELECT ONE APPLICATION

Create New Application



Create using Zia

Build a functional application effortlessly using GenAI

Create



Create from scratch

Create a new application for your business needs

Create



Create from gallery

Pick a pre-built application that suits your requirement

Pick



Import from file

Import your existing data to create an application

Import

4. ENTER APPLICATION NAME

×

Create from scratch

Application Name *

Enter your application name

☐ Enable Environment [Learn More](#)

Create Application

5.CREATED NEW APPLICATION

The screenshot displays the 'Payroll Processing System' interface. The top navigation bar includes 'Payroll Processing System', a plus icon, and tabs for 'Design', 'Workflow', and 'Settings'. The left sidebar lists navigation options: 'Dashboard', 'Employees', 'Payroll Configur...', 'Payroll Records', '+ Payroll Records' (highlighted), 'Payroll Records Re...', 'Board', and 'Pay Components'. The main content area is titled 'Pay Components' and contains the following form fields:

- Component Name: Text input field.
- Component Type: Dropdown menu with '-Select-' selected.
- Rate: Text input field with a placeholder '#####.##'.
- Is Active: Checkbox.
- Effective From: Date input field with a placeholder 'dd-MMM-yyyy'.
- Effective To: Date input field with a placeholder 'dd-MMM-yyyy'.

At the bottom of the form are two buttons: 'Submit' (purple) and 'Reset' (light gray).

6. SELECT ONE FORM

The screenshot shows a selection screen titled 'How would you like to create your form?'. It is divided into two main sections: 'On my own' and 'From a template'.

On my own

- From scratch: Button with a plus icon.
- Import Schema: Button with a schema icon.
- Using Zia: Button with a Zia icon.

From a template

- Cab Booking: Button with a car icon.
- Conf Room Booking: Button with a room icon.
- Appointment Booking: Button with a calendar icon.
- Hotel Room Booking: Button with a hotel icon.
- Complaints: Button with an exclamation mark icon.
- 50+ View More: Button with a plus icon and a link to view more templates.

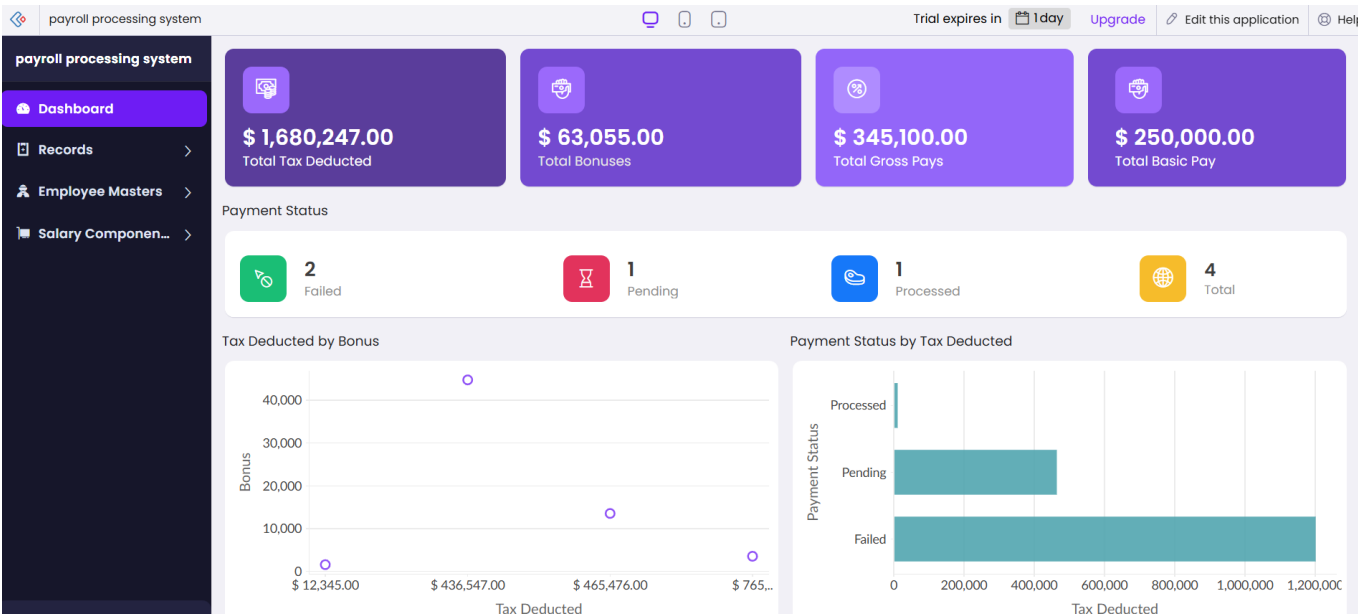
7.THE SOFTWARE HASE BEEN CREATED.

payroll processing system

payroll processing system

DashboardRecords+ Payroll RecordAll RecordsPayment StatusesEmployee MastersSalary Componen...Payroll Record

EmployeeSalary ComponentPayment DatePayment StatusBank AccountTax DeductedBonusNotesSubmitReset



Experiment 3:

AIM:

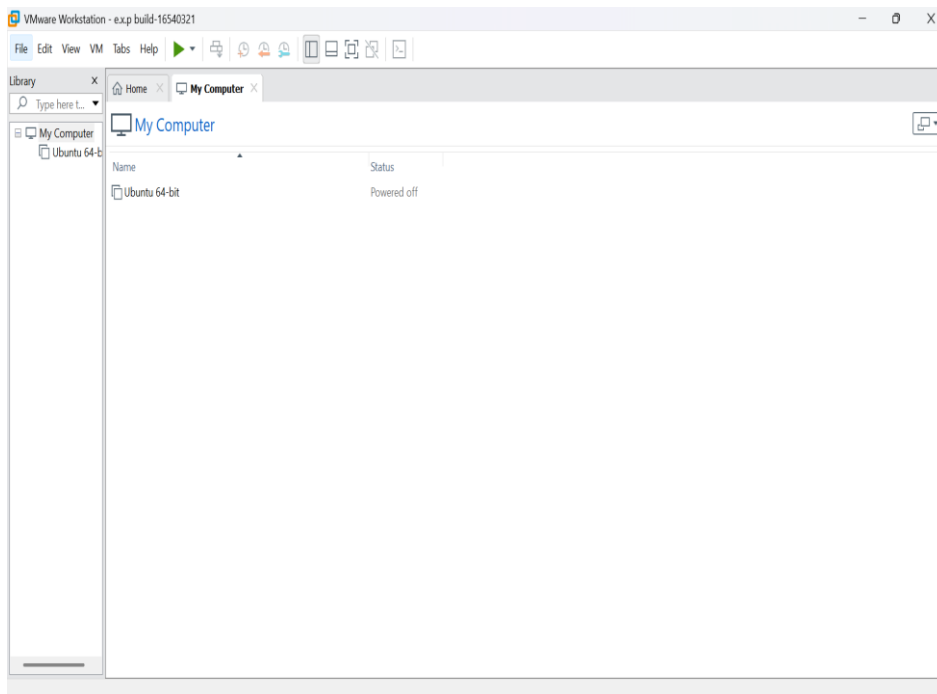
To Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Procedure:

1. Download VMware workstation and installed as type 2hypervisor.
2. Download ubuntu or tiny OS as iso image file.
3. In VMware workstation->create new VM.
4. Do the basic configuration settings.
5. Created tiny OS virtual machine.
6. Launch the VM.

IMPLEMENTATION:

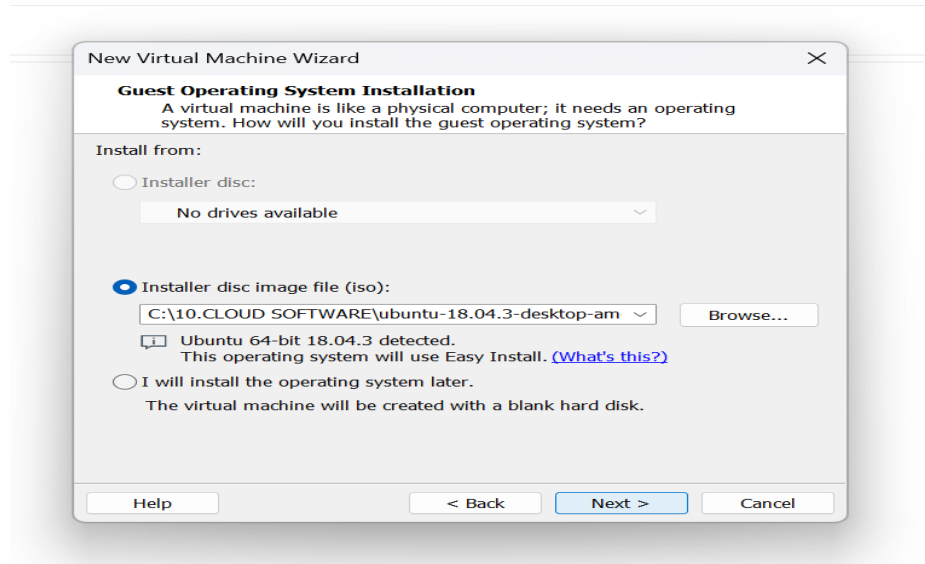
1.Download Vmware Workstation Installed As Type 2hypervisor



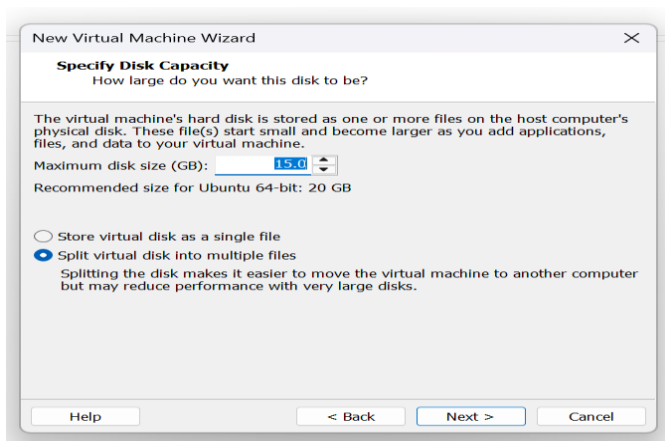
2.Download ubuntu or tiny OS as iso image file.



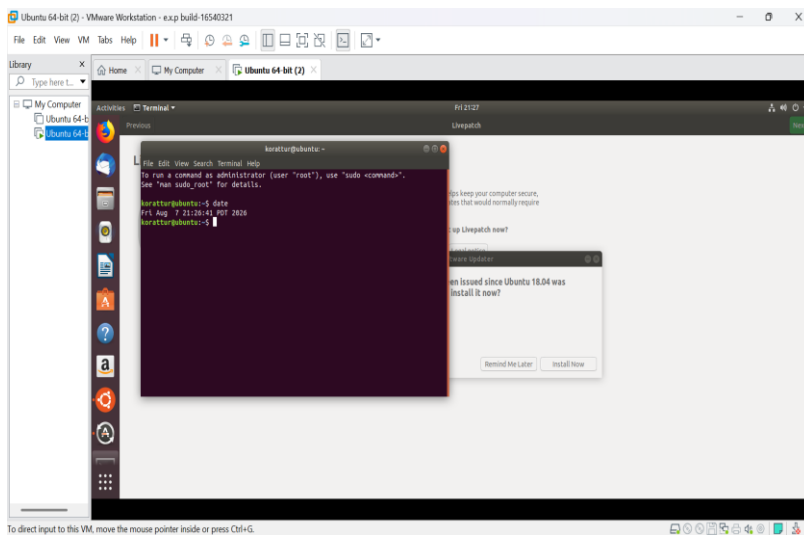
3.In VMware workstation->create new VM.



4.Do the basic configuration settings.



5.Created tiny OS virtual machine.



Result:

The Ubuntu virtual machine was successfully created using VMware Workstation.

Experiment 4:

Aim

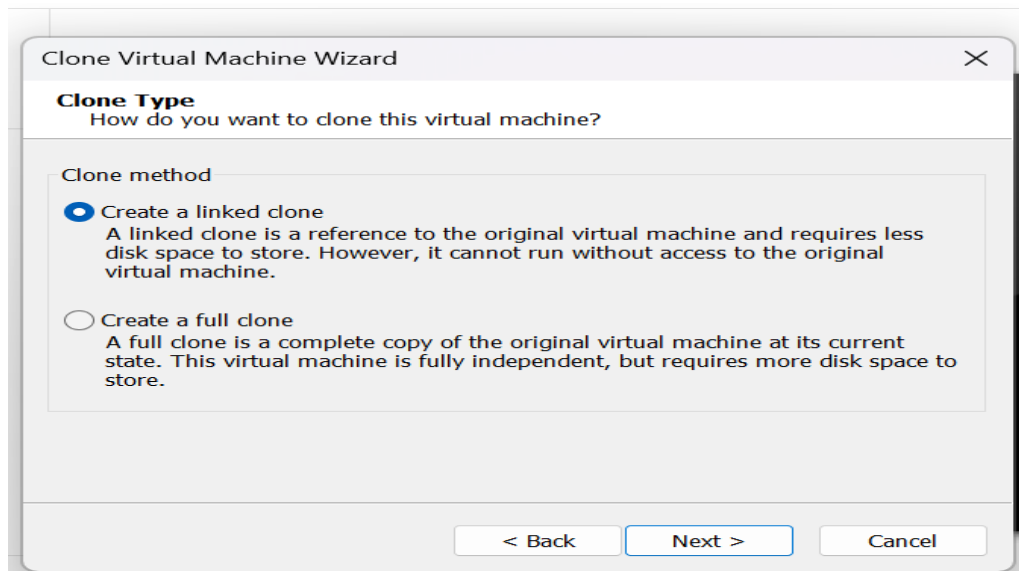
To demonstrate virtualization using VMware Workstation (Type-2 Hypervisor) by creating a virtual machine, cloning it, and testing the cloned VM.

PROCEDURE:

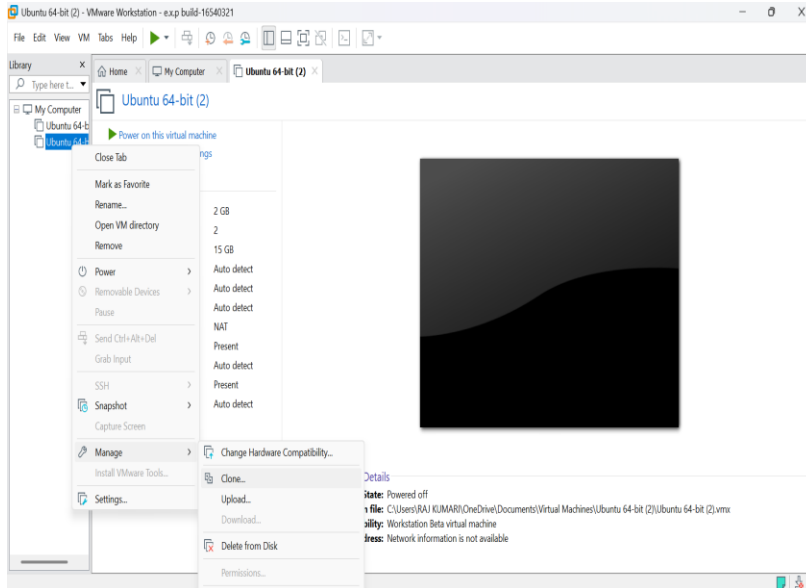
1. Download VMware workstation and installed as type 2hypervisor.
2. Download ubuntu or tiny OS as iso image file.
3. In VMware workstation->create new VM.
4. Do the basic configuration settings.
5. Created tiny OS virtual machine.
6. Launch the VM.

IMPLEMENTATION:

1. Download VMware workstation and installed as type 2hypervisor



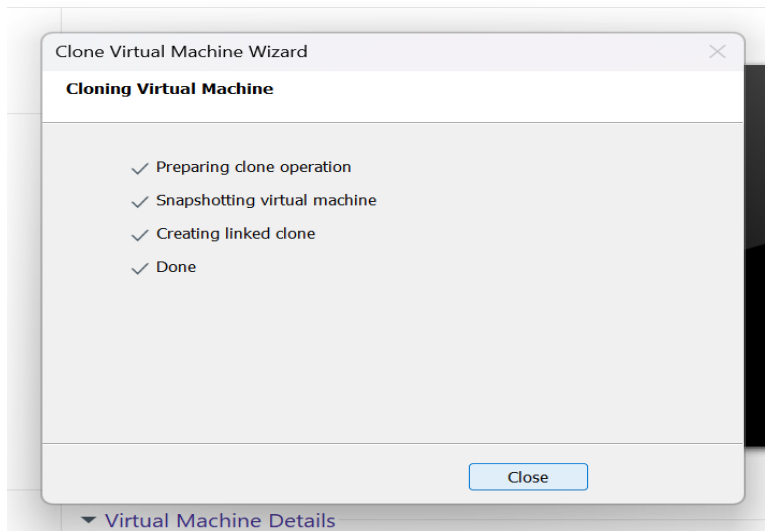
2. Download ubuntu or tiny OS as iso image file.



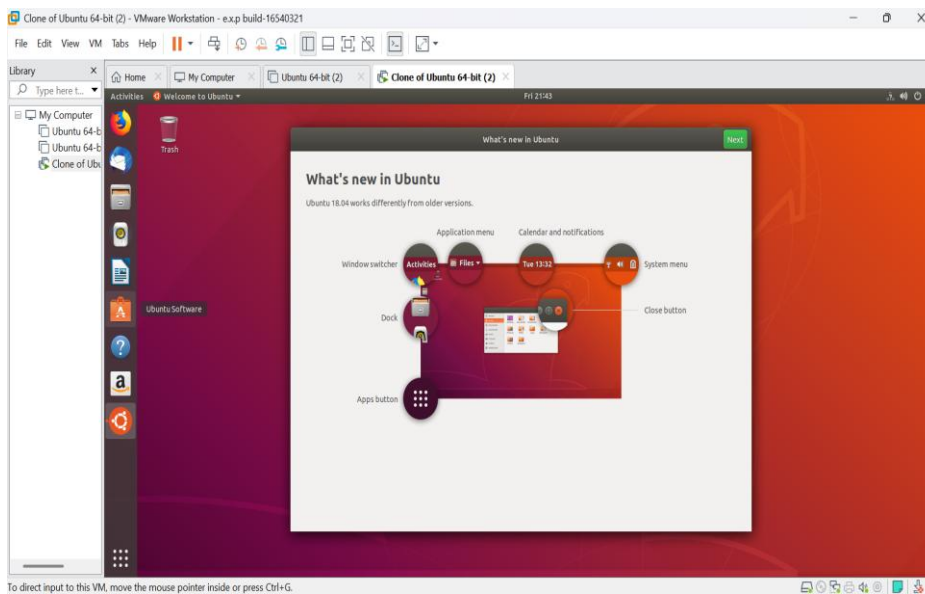
3. In VMware workstation->create new VM.



4. Do the basic configuration settings.



5. Created tiny OS virtual machine.



Result:

The Ubuntu virtual machine was successfully created using VMware Workstation.